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#include <stdio.h>

int main()
{
    int n, choice;
    printf("Enter number of processes: ");
    scanf("%d", &n);

    int at[n], bt[n], pr[n], rt[n];
    int ct[n], tat[n], wt[n];
    int pid[n];

    printf("Enter priorities (1 for higher number means higher priority, 2 for lower number means higher priority): ");
    scanf("%d", &choice);

    printf("Enter priorities:\n");
    for(int i=0;i<n;i++)
        scanf("%d",&pr[i]);

    printf("Enter arrival times:\n");
    for(int i=0;i<n;i++)
        scanf("%d",&at[i]);

    printf("Enter burst times:\n");
    for(int i=0;i<n;i++)
    {
        scanf("%d",&bt[i]);
        rt[i] = bt[i];
        pid[i] = i+1;
    }

    int time = 0, completed = 0;

    while(completed < n)
    {
        int index = -1;

        for(int i=0;i<n;i++)
        {
            if(at[i] <= time && rt[i] > 0)
            {
                if(index == -1)
                    index = i;
            }
        }
    }

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        else
        {
            if(choice == 1)
            {
                if(pr[i] > pr[index])
                    index = i;
            }
            else
            {
                if(pr[i] < pr[index])
                    index = i;
            }
        }
    }
}

if(index != -1)
{
    rt[index]--;
    time++;

    if(rt[index] == 0)
    {
        ct[index] = time;
        tat[index] = ct[index] - at[index];
        wt[index] = tat[index] - bt[index];
        completed++;
    }
}
else
{
    time++;
}
}

float avg_tat = 0, avg_wt = 0;

printf("\nPriority scheduling (Pre-Emptive):\n");
printf("PID\tPrior\tAT\tBT\tCT\tTAT\tWT\tRT\n");

for(int i=0;i<n;i++)
{
    printf("P%d\t%d\t%d\t%d\t%d\t%d\t%d\t%d\n",
        pid[i], pr[i], at[i], bt[i], ct[i], tat[i], wt[i], rt[i]);
}

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        avg_tat += tat[i];
        avg_wt += wt[i];
    }

    avg_tat /= n;
    avg_wt /= n;

    printf("\nAverage turnaround time: %.2f\n", avg_tat);
    printf("Average waiting time: %.2f\n", avg_wt);

    return 0;
}
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"C:\Users\Admin\Desktop\ × + ∨
Enter number of processes: 7
Enter priorities (1 for higher number means higher priority, 2 for lower number means higher priority): 2
Enter priorities:
3 4 4 5 2 6 1
Enter arrival times:
0 1 3 4 5 6 10
Enter burst times:
8 2 4 1 6 5 1

Priority scheduling (Pre-Emptive):
PID    Prior  AT    BT    CT    TAT    WT    RT
P1      3      0      8    15    15      7      0
P2      4      1      2    17    16    14      0
P3      4      3      4    21    18    14      0
P4      5      4      1    22    18    17      0
P5      2      5      6    12      7      1      0
P6      6      6      5    27    21    16      0
P7      1     10      1    11      1      0      0

Average turnaround time: 13.71
Average waiting time: 9.86

Process returned 0 (0x0)  execution time : 51.192 s
Press any key to continue.
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